

The NASA Sounding Rocket Program (NSRP) is a NASA run program of sounding rockets which has been operating since 1959. The missions carried out by this program are primarily used for scientific research, particularly low gravity and material based research. NASA's sounding rocket program is commonly used by colleges and universities for upper atmosphere research.

## Program

In 1965, NASA's cost of a sounding rocket system was \$5,000 to \$150,000, using combinations of stage motors from the Aerobee, Hercules M5E1 (developed for the Nike Ajax), and Thiokol Apache.

The program was consolidated at the Wallops Flight Facility in the 1980s and uses extra military solid rocket motors. Rockets are frequently launched from fixed facilities at Wallops, the Navy's White Sands Missile Range in New Mexico, the Poker Flat Research Range in Alaska, Kwajalein, Marshall Islands, Pacific Missile Range Facility in Barking Sands, Hawaii, and Andøya Rocket Range, Norway. The rockets are categorized as "Significant Military Equipment" for ITAR.

## Propulsion

Rockets in use include single-stage or combinations of:

Black Brant family, 17.26" diameter

Improved Orion surplus motor, 14" diameter

Terrier/Hercules MK12 or MK70 surplus motors, 18" diameter

Oriole motor

Magellan Aerospace Nihka exo-atmospheric motor, 17.26" diameter, 192,878 pound-seconds impulse

Thiokol Improved Malemute TU-758 surplus motor, 16" diameter

Hercules Talos surplus motor, 30.1" diameter

Some combinations of stages allow payloads of up to 1550 pounds.

## Launches

### PolarNOx

The PolarNOx mission was a set of experimental launches used to measure the nitric oxide present in the upper atmosphere that is produced by auroras.

### DEUCE

The DEUCE (Dual-channel Extreme Ultraviolet Continuum Experiment) mission was planned to obtain scientific data about the IGM. This failed however due to problems with the attitude control system.

On 11 July 2022 a Black Brant IX rocket from Arnhem Space Centre launched the fourth DEUCE ultraviolet astronomy mission following flights in 2017, 2018, and 2020 for NASA (CU Boulder). The suborbital flight had apogee of 162 mi (261 km) and was successful.

### ASPIRE

The ASPIRE mission (Advanced Supersonic Parachute Inflation Research Experiment) was an experiment which tested a Mars mission parachute design. The mission consisted of three tests using the Black Brant IX sounding rocket, with the third and final test taking place on Sept. 7, 2018.

### AZURE

The Auroral Zone Upwelling Rocket Experiment in April 2019 caught many Norwegians by surprise by triggering an unusual form of the Aurora Borealis. [1]

== References ==